

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence
Topic: PEAS Framework and Task Environment Analysis

Objectives

The goal of this assignment is to apply the formal frameworks from *Russell & Norvig (Chapter 2)* to a real-world problem. You will demonstrate your ability to specify success metrics and classify the complexity of an AI's operational environment.

Task Description

Select **one** of the following modern AI applications (or propose one for approval):

- **A:** An Automated Warehouse Sorting Robot (e.g., Amazon Robotics).
- **B:** A Personalized News Feed Recommender (e.g., Twitter/X or TikTok algorithm).
- **C:** An Agricultural Monitoring Drone for crop disease detection.

For your chosen system, complete the following three tasks:

Task 1: PEAS Specification

Define the **PEAS** framework for your agent. Ensure your descriptions are specific (e.g., do not just say "Cameras," specify "High-resolution RGB and Infrared sensors").

Performance Measure: How exactly is success measured? List at least four metrics.

Environment: Where does the agent act? What are the key external factors?

Actuators: What physical or digital tools does the agent use to change its world?

Sensors: What are the inputs the agent uses to perceive its state?

Task 2: Environment Classification

Classify the task environment according to the following dimensions. For each, provide a **one-sentence justification** explaining your choice.

- Fully Observable vs. Partially Observable
- Single-agent vs. Multi-agent
- Deterministic vs. Stochastic
- Episodic vs. Sequential
- Static vs. Dynamic
- Discrete vs. Continuous

Task 3: Critical Analysis

Answer the following questions in short-paragraph format:

1. Which of the six dimensions mentioned in Task 2 poses the **greatest technical challenge** for an AI engineer building this system? Why?
2. Propose one **Utility Function** that would help the agent handle a trade-off (e.g., Speed vs. Safety). How would the agent's behavior change if the weight of this utility was doubled?

Submission Guidelines

- Submit your response as a single **PDF** file via email.
- Format: 12pt font, maximum 2 pages.
- Academic Integrity: All justifications must be your own. Use of AI for formatting is allowed, but the design logic must be original.